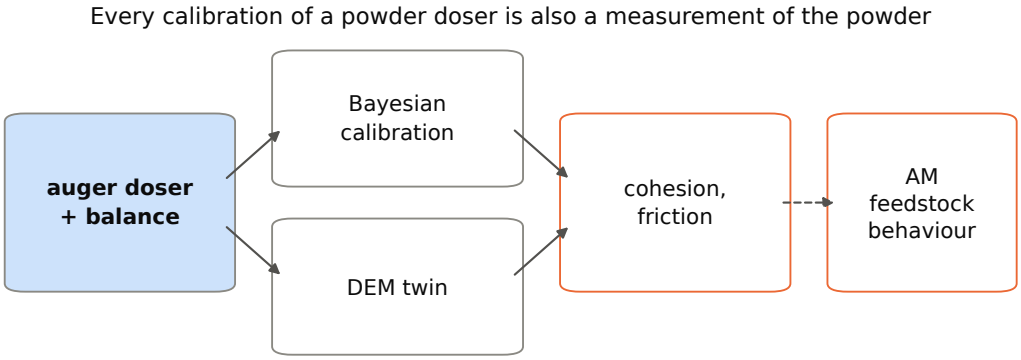
Graphical Abstract

Auger-based powder dosing as a probe of powder flow: multi-objective, multi-task Bayesian calibration and physics-based property inference

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Highlights

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- Calibration of an open-source auger doser is cast as Bayesian optimization
- Optimized settings dosed salt three times faster than hand-tuned settings
- Delivery per auger revolution spans a 24-fold range across 13 powders
- Delivery ranks 12 powders like their literature angle of repose
- A DEM model, not fitted to dosing data, matches salt delivery at one setting

Auger-based powder dosing as a probe of powder flow: multi-objective, multi-task Bayesian calibration and physics-based property inference

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Abstract

Every powder in an alloy-discovery workflow must be calibrated on its doser, and because delivery depends on both the drive settings and the powder, each calibration is also a small measurement of powder flow. We treat calibration of an open-source, 3D-printed auger doser as multi-objective Bayesian optimization and ask what the calibration data reveal about the powder. At fixed settings, 13 powders delivered between 10 and 231 mg per auger revolution, and three delivered none; across 12 of them, delivery ranked like the literature angle of repose (Spearman $\rho = -0.84$). On table salt, an optimizer tuned eight controller settings in an unattended 42-dose campaign. Its recommended settings dosed 0.5 g in a median 78 s with a median error of 1.8 mg over eight repeats, three times faster than hand-tuned settings, and dosed 8 g of AlSi10Mg to within 0.1% unchanged. Each powder then becomes a related task of one multi-task model, with added objectives for repeatability and accessible dose range. A discrete element model of the auger, with no parameters fitted to dosing data, reproduces the measured delivery of salt at the campaign's centre point (105 versus 105 $\pm$ 11 mg per revolution) and will infer an effective cohesion and friction for each powder from its dosing response. [Multi-task calibration cut the doses needed for a new powder by X%, and the inferred parameters tracked shear-cell and Hall-flow measurements.]

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Because every dosing workflow calibrates anyway, those records can double as a feedstock characterization at little extra cost.

Keywords: powder dosing, powder flowability, Bayesian optimization, multi-task learning, discrete element method, self-driving laboratory

1. Introduction

Alloy discovery by additive manufacturing (AM) starts with weighing powders. Each candidate composition is made by blending elemental or pre-alloyed powders, either in a bed for laser powder bed fusion (L-PBF) or in the feeders of a directed energy deposition (DED) system [1, 2, 3, 4, 5, 6]. In a self-driving laboratory (SDL), where a planning algorithm chooses each next composition and robots make and test it [7, 8, 9], that weighing step must be automatic, accurate and fast for every powder the workflow uses [10, 11, 12, 13]. The same powders must later spread into thin, uniform layers, and how well they do so depends on the same properties that make them easy or hard to dose: particle size and shape, cohesion and friction [14, 15, 16, 17].

Measuring those properties is slow and often uninformative. Hall and Carney funnels report a single flow time and cannot test strongly cohesive powders, which do not flow through them at all [18, 19]. Shear cells and powder rheometers give richer data, but each test needs tens of millilitres of powder, a dedicated instrument and a trained operator [20, 21]. Rotating drums and spreadability tests come closer to the conditions inside an AM machine [22, 23, 24, 25], yet different flow tests often rank the same powders differently, and none of them predicts spreading reliably on its own [18, 19, 24]. For an SDL that cycles through dozens of feedstocks, including small batches of reactive or expensive powders, a separate characterization campaign for each powder is rarely affordable.

Powder feeders, by contrast, are characterized all the time, because each new powder must be calibrated before it can be dosed. In pharmaceutical manufacturing, the feed factor of a screw feeder, the mass it delivers per screw revolution, has been predicted from bulk density, compressibility and shear-cell data [26, 27, 28], and loss-in-weight feeders have standard methods for measuring how their output varies [29, 30, 31]. If material properties predict feeder behaviour, the reverse should also hold: a feeder that is driven through a designed set of conditions, and weighed, measures the powder. Small-scale dosers make this attractive because the measurement uses milligrams to

grams of powder and happens anyway, every time a new powder is loaded [32, 33, 34, 35, 36].

Calibration of this kind is an optimization problem with several competing goals. A dose should be accurate and repeatable, it should be fast, and the settings should work across the range of doses the workflow needs. Overshoot cannot be undone, because powder cannot be taken back out of the cup, so an error in one direction costs more than in the other. Bayesian optimization (BO) suits such problems. It fits a probabilistic model to the doses made so far and chooses the next settings with an acquisition function, a rule that scores candidate settings by their expected gain [37, 38, 39]; it has already tuned AM processes autonomously [40]. Multi-objective BO finds the trade-off between goals such as time and error, the Pareto front [41], and priors that favour a few influential settings make it practical with many settings and few experiments [42].

Each new powder, however, still starts from scratch. Multi-task BO treats related problems, here different powders on the same doser, as tasks of one model, so that data from earlier powders inform the first doses of a new one [43, 44, 45]. How strongly two powders are related is then a quantity the model learns, and it is natural to ask whether it reflects their physical properties.

Physics-based models can answer that question directly. The discrete element method (DEM) simulates every grain and every contact between grains [46, 47]. It has been used to predict the output of screw conveyors [48], and it represents cohesion and rolling friction with contact laws whose parameters are hard to measure directly [49, 50, 51]. Those parameters are usually calibrated by matching simulations to bulk tests, which is itself an inverse problem [52, 53, 54, 55] and is best treated with the statistical methods developed for calibrating computer models [56, 57]. The closest precedent to the present work calibrated DEM parameters of pharmaceutical powders against several bulk responses extracted from auger dosing experiments [58, 59]. Here the same idea is applied to an instrument that also optimizes its own settings, and to AM feedstocks. DEM is also the standard tool for studying how powders spread into AM layers [60, 61, 15, 62]. An auger turned at controlled speeds and tilts is a well-defined shear and discharge test, so a DEM model of the auger could convert dosing data into contact parameters that a spreading model can then use.

In this work we build on an open-source, 3D-printed auger doser [63] and make three contributions. First, we formulate per-powder calibration

as multi-objective BO over the doser's controller settings and demonstrate it on sodium chloride, where a search over eight settings ran unattended (Section 3.2). Second, we show that the doser's response at fixed settings separates 13 powders over a 24-fold range, and we extend calibration to many powders as a multi-task problem whose objectives also include repeatability and the range of doses a powder can reach (Sections 3.1 and 3.3). Third, we build a DEM model of the auger, check it against measured delivery without fitting any parameter to dosing data, and use it to infer effective cohesion and friction from dosing data, which we compare with Hall-flow and shear-cell measurements (Sections 3.4 and 3.5). All hardware, firmware, optimization code, simulation inputs and data are open [64].

2. Materials and methods

Sections 2.1 to 2.3 describe the doser, the powders, and the fixed-setting tests from which each powder's dosing fingerprint is taken. Sections 2.4 and 2.5 describe the calibration campaign on salt and its planned extension to many powders. Sections 2.6 to 2.8 describe the discrete element model, the planned characterization of the same powder lots, and the statistics.

2.1. Doser and dose procedure

The doser is one channel of an open-source design described in detail elsewhere [63, 64]; this section summarizes the parts that matter here. A one-piece auger, printed in PLA on a Bambu Lab H2D printer with 0.2 mm layers, serves as both reservoir and screw. Its tube, 21 mm in bore and 250 mm long, holds up to 58 mL of powder, and a single helical flight (0.5 mm thick, 10.4 mm pitch, 8 turns) runs over the last 84 mm before a 12 mm conical outlet with a 3 mm exit hole. A NEMA-11 stepper motor (11HS18-0674S, Pololu Tic T500 driver) turns the auger through a 20:44 gear pair at up to about 109 rpm. Two servos tilt the auger about a hinge near the outlet through a 2:1 gear, and a push–pull solenoid (JF-0530B, 60 ms pulses) taps the tube to free powder held at the lip of the outlet. Powder falls into a cup on an analytical balance (A&D HR-100A; 0.1 mg readability, 102 g capacity) that reports each reading over a serial (RS-232) link. A Raspberry Pi Pico W runs the dosing firmware (MicroPython), and a Raspberry Pi Zero 2 W next to the rig runs each dose, records it, and uploads the record to a database. All tilts below are angles of the auger axis above horizontal.

A dose has three stages (Fig. 1a). In the *bulk* stage the auger turns continuously at a set tilt and speed, optionally while the solenoid taps at 2 Hz, until the mass still to go falls to a switch threshold plus a fixed 50 mg allowance for powder that is still falling. In the *trickle* stage the auger runs slowly (typically 2–15 rpm) under a proportional–integral controller on the flow rate, which a Kalman filter estimates from the noisy balance readings. The trickle stops when the predicted final mass comes within a fixed 35 mg margin of the target; the prediction is the current mass estimate, plus the afterflow still expected (the flow rate times an afterflow time constant τ_{af}), plus one standard deviation of the estimate. In the *tap* stage the auger stops, the tube moves to a third tilt, and single taps free a few milligrams at a time until the reading lies within a stopping band of $\pm\delta$ around the target. The dose is scored on a balance reading taken 2 s after the last action.

Two rules end a dose early: a dose that exceeds the target by more than 100 mg is aborted, and a dose that uses up its allowance of taps ends as a jam. The Kalman filter models the balance as a first-order lag, that is, a reading that approaches the true mass with a single time constant. The controller used the datasheet value of 0.7 s for that constant, whereas drop tests on the rig gave 0.16 s. The optimized settings absorb this mismatch, but it should be removed before settings are compared across powders. Every dose writes one machine-readable record with the settings as executed, the time spent in each stage, every stop event, and the balance readings at 4 Hz.

2.2. Powders

We report dosing data for 13 powders (Table 1). Seven are food-safe stand-ins chosen to span a range of flow behaviour: table salt, calcium lactate, xanthan gum, carboxymethyl cellulose, white and brown rice flour, and sodium alginate. Six are relevant to alloy and materials work: gas-atomized AlSi10Mg, silicon sieved to 75–150 µm and to below 44 µm, sodium sulfate, barium chloride dihydrate, and fumed silica. An Al 4047 powder made in our laboratory by ultrasonic atomization was also dosed. Particle-size distributions, densities and flow properties have not yet been measured on these lots. For the correlation analysis in Section 3.5 we therefore use literature and supplier values for the material class, each traced to its source in the repository [64]; seven of the 12 angles of repose are engineering estimates. [Measured properties of each lot (Section 2.7) will replace these values.]

Table 1: Powders and their response to the auger at fixed settings (protocol C: 30 rpm, six single revolutions per tilt). Delivery is the mean mass per auger revolution ± standard error at 22.5° tilt. Closed-loop doses used an earlier three-stage controller with one set of settings tuned by hand on salt. "–" marks powders that delivered no measurable amount.

Powder	Group	mg/rev at 22.5°	1 g doses within ±5%
AlSi10Mg (gas-atomized)	research	231.0 ± 12.7	3/3
Silicon, 75–150 µm	research	210.7 ± 6.0	3/3
Sodium sulfate	research	208.4 ± 13.1	3/3
Calcium lactate	stand-in	198.3 ± 1.8	3/3
Barium chloride dihydrate	research	184.7 ± 12.9	not dosed (caked)
Xanthan gum	stand-in	161.2 ± 7.9	3/3
Sodium chloride (table salt)	stand-in	146.5 ± 11.0	6/6
Carboxymethyl cellulose	stand-in	26.4 ± 1.0	3/3
White rice flour	stand-in	12.8 ± 2.0	0/3 (short)
Sodium alginate	stand-in	9.6 ± 0.3	0/3 (short)
Silicon, <44 µm	research	–	0/6
Brown rice flour	stand-in	–	0/3
Fumed silica	research	–	not dosed

2.3. Gravimetric response at fixed settings

Each powder was first run through a fixed set of test protocols, described fully with the hardware [63]. Four of them are used here. Protocol C turns the auger through six single revolutions at 30 rpm at each of three tilts (0, 22.5 and 45°), with a settled balance reading after each revolution. Protocol D turns it through three revolutions at each of 15, 45 and 90 rpm at 22.5°. Protocol E applies single taps after a re-feed revolution. Protocol G makes three closed-loop doses of 1 g with an earlier, simpler three-stage controller whose settings were tuned by hand on salt and then frozen for every powder. From these records we extract a short *dosing fingerprint* for each powder (Table 2): the delivery per revolution at the three tilts, its scatter from revolution to revolution, the exponent α in $\dot{m} \propto N^{\alpha}$ for auger speed N, the mass freed by a single tap, and the afterflow time constant τ_{af} fitted from the mass that arrives after each stop. The full set of protocols uses 4–10 g of powder and takes 40–50 min. Two further descriptors were proposed for screening atomized batches [64]. The first is a gravity-assisted volumetric flow index: the delivery per revolution at 45°, divided by bulk density and normalized to a reference AlSi10Mg powder. The second is an interruption

fraction: the share of balance increments at 15 rpm that show no flow, a doser analogue of the cohesive index measured in a rotating drum [65, 22].

2.4. Calibration as multi-objective Bayesian optimization

For one powder and one target mass, calibration searches eight controller settings (Table 3) to minimize two objectives at once: the total time of a dose, t, and its absolute error, $|e|$, both measured on the scored balance reading. The campaign fixed no tolerance in advance. The stopping band δ is itself one of the searched settings, and the error objective measures where the dose actually ended. Doses slower than 180 s or more than 20 mg off count as failures. These two limits form the reference point for the hypervolume, the area of objective space between that reference point and the best doses so far, which is the usual single-number summary of a Pareto front. Each dose ends in one of a fixed set of outcomes: *ok* (within the band), *overshoot* (above the band), *jam* (the allowance of taps, or of the small 5° auger nudges that re-feed the lip in the tap stage, used up, or no progress for 8 s), *timeout* (600 s), *abort* (100 mg above the target), or an equipment error (balance or communication), which is flagged separately. Spills are flagged by an operator and were therefore not recorded in unattended runs. All other controller parameters, including the controller gains and the 100 mg abort limit, were frozen at the values of the hand-tuned controller.

The campaign ran in three phases, with four doses at the hand-tuned settings spread through it as a baseline and as a check on drift. In the first phase, a two-level fractional factorial screening design of resolution IV (2^{8-4}, 16 runs) estimated the main effect of each setting without confusing it with any two-factor interaction; each run sits at a corner of the search space, with every setting at its high or low level. Four repeated centre-point doses, with every setting at the middle of its range, measured the dose-to-dose scatter. In the second phase, the afterflow constant τ_{af} was refitted from the 26 stop events recorded so far (a median-ratio estimate, which resists outliers; $\tau_{\mathrm{af}} = 0.834$ s), and the centre point and the hand-tuned settings were repeated with the new value. In the third phase, 14 doses were chosen one at a time in Ax 0.4.3 [66] on BoTorch [67]: two quasi-random (Sobol) doses, then 12 chosen by the optimizer.

The optimizer's model was a fully Bayesian Gaussian process with a sparse axis-aligned subspace prior (SAASBO) [42], which assumes that only a few of the eight settings matter strongly and therefore suits a search with many settings and few observations. Its acquisition function was the logarithmic

form of the noisy expected hypervolume improvement (qLogNEHVI) [41, 68], which allows for the noise in each observed dose. The screening and centre-point doses were given to the model as existing data, so that it used all 28 earlier doses. Doses that jammed were reported to the model at the failure limits (180 s, 20 mg), and overshoots at their measured values. Each suggestion took 1–1.5 min to compute. A preset deadline stopped the optimization phase after 14 doses, short of the 30–40 planned.

Each dose ran unattended over a remote connection to the rig, with the cup emptied at fixed intervals. A restart never repeated a dose, because each dose carries a unique identifier and its stored result is returned instead. An AI coding agent (Claude Code, Anthropic) launched and monitored the campaign and the validation blocks; it also wrote much of the firmware, campaign, analysis and simulation code, which the authors reviewed. The agent started the campaign script, watched it, and restarted it after interruptions, but the settings of every dose came from the screening design or the optimizer, not from the agent. Every prompt and response is public in the repository [64].

Because the observed Pareto front is built from single doses, a point can lie on it by chance. We therefore recommend settings from the model's own Pareto set, computed from its averaged (posterior mean) predictions rather than from the observed doses, and validate them with a block of eight repeated doses on a later day. The validation block also tested the screening run that lay on the observed front (`corner-09`).

2.5. Multi-task calibration across powders

Each powder is a task, and all tasks share the eight controller settings and the dose record. A multi-task Gaussian process [43, 44] models the objectives of task k at settings $\mathbf{x}$ with a product kernel $K\Big((\mathbf{x}, k), (\mathbf{x}', k')\Big) = B_{kk'}\, k_x(\mathbf{x}, \mathbf{x}')$, in which the positive semi-definite matrix B is learned from the data and its normalized entries are the correlations between powders. We will compare two variants, both available in the version of Ax used here. An index kernel learns B freely from the dose data. A descriptor kernel computes B from each powder's dosing fingerprint (Section 2.3) and, where available, its measured properties, so that a new powder can borrow from similar ones before its first optimized dose [45].

The objectives are extended from two to four, following the formulation developed for this controller [64]: (i) accuracy, the median absolute error; (ii) repeatability, the probability that a dose lands within tolerance, $P(|e| \leq \delta)$, estimated from blocks of repeated doses, together with a constraint on the

probability of overshoot, which cannot be undone; (iii) time per dose; and (iv) the accessible dose range, the smallest target that the settings deliver within tolerance with probability at least 0.95. [Final formulation (constraint thresholds, replicate count per evaluation, and dose targets) to be fixed before the multi-task campaigns.]

The benchmark has two parts. In the retrospective part, each completed single-task campaign is held out in turn, and a statistical model fitted to its doses (a surrogate) stands in for the rig. Multi-task optimization on the held-out powder, informed by the other powders, is then compared with single-task SAASBO, with quasi-random (Sobol) search, and with simply carrying over the best settings of the most similar powder. In the prospective part, one new powder, a freshly atomized batch, is calibrated with the multi-task model. The figure of merit is the number of doses until the recommended settings reach 90% of the best hypervolume known for that powder. [Planned campaigns: AlSi10Mg, 316L stainless steel, sieved Al 4047 and at least one elemental alloy precursor, each with about 42 doses and an 8-dose validation block.] Reactive elemental precursors will be dosed in an argon glove box; where the elemental form is too hazardous to handle as a fine powder, a master alloy will be used instead.

2.6. Discrete element model of the auger

A discrete element model of the auger outlet was built in LIGGGHTS-PUBLIC 3.8 [69, 46]. Grains interact through a Hertz normal contact with history-dependent tangential friction and an elastic–plastic spring–dashpot rolling resistance model [51]. Cohesion is added as a simplified Johnson–Kendall–Roberts (SJKR) term, a pull-off force proportional to the cohesion energy density and the contact area [49]. For salt, the grains follow a three-size distribution around a median diameter of 0.425 mm (number fractions 0.3/0.4/0.3 at 0.85, 1 and 1.15 times the median) with a density of 2165 $\mathrm{kg\,m^{-3}}$, a literature value for the material class. Young's modulus is reduced to 0.2 MPa to allow a time step of 2.1×10^{-5} s, which keeps mean overlaps below 0.4% of a diameter; Poisson's ratio is 0.3, the coefficient of restitution 0.5, sliding friction 0.5 between grains and 0.4 against the wall, and rolling friction 0.3. None of these values was fitted to the dosing data. The geometry is meshed from the CAD model of the printed auger: the conical outlet with its 3 mm exit, the solid core whose tip leaves a 1.07 mm annular gap at the exit, and the lowest 1.5 turns of the flight, filled with about 29,000 grains and topped up from a feed zone. Tilt is applied by rotating gravity, rotation by

turning the auger mesh, and the delivered mass is counted as grains leave the exit. Each case was run for 0.3–3 revolutions, mostly from a single random packing. Two caveats follow from how the model is built. Because the SJKR pull-off force scales with the contact area, which depends on the softened stiffness, any cohesion energy density inferred with this model is specific to the simulator and its numerical stiffness, and comparable across powders only at that fixed stiffness. And adhesion between particles and the printed PLA wall is a separate parameter that dosing data alone may not resolve.

To infer effective contact parameters for a powder, the model will be run over a designed set of cohesion energy densities and sliding and rolling friction coefficients, with the particle-size distribution fixed at the measured one, at the tilts and speeds of the dosing fingerprint. A Gaussian-process emulator, a fast statistical stand-in fitted to these runs, then replaces the simulator in a Bayesian calibration [56, 57]. The calibration returns a posterior distribution over the contact parameters, that is, the range of parameter values consistent with the measured fingerprint and its scatter, with a discrepancy term for systematic model error. Dosing data alone may not separate cohesion from friction [52, 53, 59], so whether each parameter can be recovered on its own will first be checked on synthetic data before the inference is run on measured data. For the finer metal powders, the grain count will be reduced by coarse-graining, which simulates fewer, larger grains while preserving the ratio of cohesive to gravitational forces [70]. [Emulator design (number of runs, parameter ranges) to be fixed after the friction and cohesion sweeps in the rig geometry.]

2.7. Independent powder characterization

To test whether the dosing-derived descriptors and inferred parameters carry physical meaning, the same lots will be characterized by standard methods [71]: particle-size distribution by laser diffraction [72] or sieving; bulk and tapped density and the Hausner ratio [73]; Hall-flow time [74] or, for powders that do not pass the Hall funnel, Carney-flow time [75]; angle of repose; and the flow function coefficient ff_c from a shear cell [76, 20]. Spreadability will be assessed by blade-spreading single layers of 50–100 µm and measuring their areal mass and coverage [24, 25]. [Shear-cell measurements are expected to be outsourced; sieves have been ordered.]

2.8. Statistical analysis

Delivery per revolution is reported as the mean ± standard error of single revolutions, and a value counts as a measurement only when the mean exceeds twice its standard error. Associations between dosing descriptors and powder properties are reported as Spearman rank correlations, because $n \leq 12$ and many property values are class estimates; the p-values are not corrected for the 42 predictor–response pairs examined and are treated as leads rather than findings. The optimization models are specified completely by the fixed software versions and the campaign configuration stored with each campaign, and all trials, including aborted and jammed doses, are kept in the analysis. [For the multi-task and inference results, report baselines, held-out performance, residuals and hyperparameters as required by the journal's guidelines for machine-learning studies.]

3. Results

Sections 3.1 and 3.2 report measured results: the response of 13 powders at fixed settings, and the optimization campaign on salt. Section 3.3 shows the intended form of the multi-task results, which do not exist yet. Section 3.4 reports the discrete element model and Section 3.5 the comparison with powder properties; within those two sections, the inference and the same-lot comparisons are placeholders.

3.1. Dosing response separates powders

At fixed settings the auger delivered very different amounts of different powders (Fig. 2a, Table 1). Ten of the 13 powders delivered between 9.6 mg (sodium alginate) and 231 mg (AlSi10Mg) per revolution at 22.5°, a 24-fold range, and three delivered no measurable amount: fumed silica, brown rice flour, and silicon finer than 44 µm. The two silicon grades are the clearest case. They are the same material, yet the 75–150 µm grade delivered 211 mg per revolution and the finer grade delivered nothing in 17 of 18 revolutions, so particle size alone switched the flow on and off.

Across all powders and tilts, powder identity accounted for 70% of the variance of the logarithm of the mass per revolution, tilt for 20%, their interaction for 4%, and repeat-to-repeat scatter for 6%. Tilting the auger from 0 to 22.5° raised delivery by a median factor of 4.2, and tilting further to 45° by only 1.2 more; carboxymethyl cellulose was the only powder that delivered less at 45° than at 22.5°. Delivery rose more slowly than auger

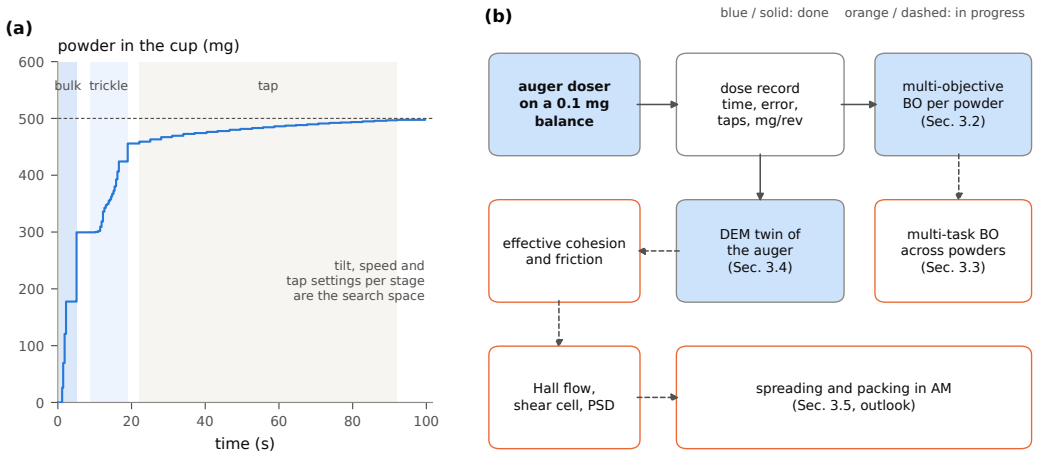


Figure 1: Calibration of an auger doser as a measurement of the powder. (a) One 0.5 g dose of salt at the recommended settings, as the balance recorded it: a fast bulk stage, a slow trickle stage and a tap stage, each with its own tilt, speed and tap settings. (b) The data flow of this work. Dose records feed a multi-objective optimizer for each powder and a multi-task model across powders. A discrete element model of the auger converts the same records into effective contact parameters, which are compared with standard flow tests and, in future work, with spreading in additive manufacturing. Solid boxes and arrows mark completed work; orange outlines and dashed arrows mark work in progress.

speed for most powders (median exponent $\alpha = 0.78$), but faster than speed for the three slowest (white rice flour 1.36, carboxymethyl cellulose 1.13, sodium alginate 1.08). Put another way, free-flowing powders delivered less per revolution as the auger sped up (salt by 33–47%, AlSi10Mg by 49% over a sixfold range of speed), whereas cohesive ones delivered more (white rice flour 94% more, carboxymethyl cellulose 23% more), so the sign of the speed response alone separates the two groups. A single tap freed between 0.24 mg (sodium alginate) and 20 mg (calcium lactate), and the afterflow time constant ranged from 0.78 to 1.17 s across six powders. Each of these numbers comes from data that any calibration of the doser collects anyway.

The same powders also differ in how well a fixed controller doses them (Fig. 2b). With an earlier three-stage controller and one set of settings tuned by hand on salt, seven powders landed within ±5% of a 1 g target on every dose (24 of 24), and the three alloy-relevant ones (AlSi10Mg, coarse silicon and sodium sulfate) within 1%. The two slowest-flowing powders ran out of time: white rice flour delivered 0.84–0.89 g and sodium alginate 0.69–0.73 g. Even among the powders that met the limit, the median time for 1 g ranged

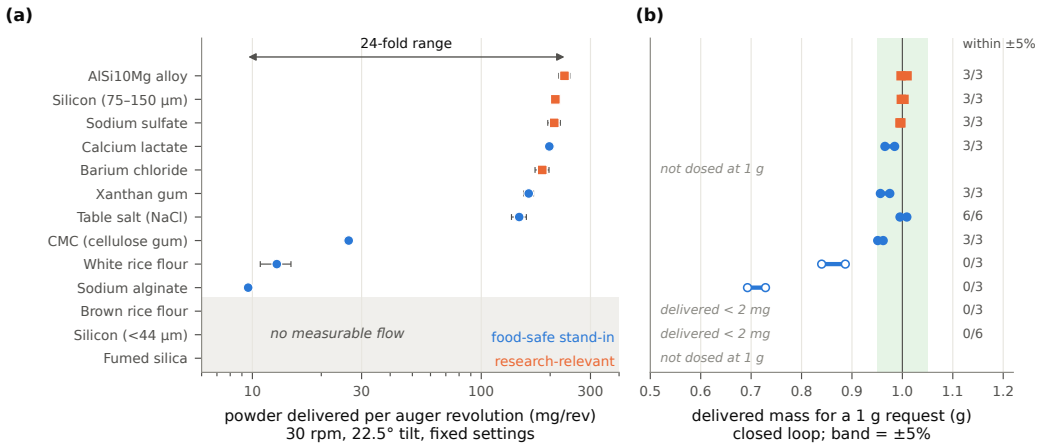


Figure 2: Thirteen powders on one auger. (a) Mass delivered per auger revolution at fixed settings (30 rpm, 22.5° tilt; mean ± standard error of six revolutions). Three powders delivered no measurable amount. (b) Delivered mass for 1 g requests under closed-loop control with one set of hand-tuned settings; bars span the replicate doses, open markers show powders that missed the ±5% band, and the numbers give doses within the band. Data from the companion hardware study [63].

from 65 s for AlSi10Mg to 11 min for carboxymethyl cellulose. A controller tuned for one powder is therefore accurate on similar powders but slow or short on different ones, which is the argument for calibrating the controller on each powder.

3.2. Bayesian calibration on salt

The salt campaign ran 42 doses of 0.5 g unattended in less than 2.5 h (Fig. 3). The four doses at the hand-tuned settings took 180–343 s (median 245 s) and ended 3.6–4.8 mg short.

The screening design showed which settings mattered and why. Tapping during the bulk stage cut the average time of successful doses from 186 to 79 s, but five of the eight runs with bulk taps on ended in an overshoot or a jam, against two of the eight with taps off, presumably because taps free extra powder after the auger stops. Raising the tap-stage tilt from 0 to 15° cut the average time from 223 to 92 s. Switching to the trickle stage earlier, with 0.30 g rather than 0.05 g to go, cut the mean absolute error from 34 to 11 mg, and two runs at the larger threshold ended in an overshoot or a jam, against five at the smaller. In total, 8 of the 42 doses overshot, by 21–83 mg. Six of those were screening runs, and every one was at the highest bulk speed,

the smallest switch threshold, or both; three more doses ended in a jam when the allowance of taps ran out.

The optimizer then chose 14 doses before the campaign reached its deadline. None of them beat the best single doses of the screening design, so the observed hypervolume did not grow during this phase. The settings the model recommended (`bo-005`, Table 3) had been predicted to take 91 ± 66 s with a 3 ± 9 mg error, and took 100 s with a 2.4 mg error in the campaign. Repeated eight times a week later, they dosed 0.5 g in a median 78 s (range 60–114 s; bootstrap 95% interval of the median 69–110 s) with a median error of 1.8 mg (1.3–2.3 mg; every dose 0.8–2.6 mg under the target) and no overshoot (Fig. 3b); the delivered masses had a standard deviation of 0.56 mg. That is three times faster than the hand-tuned settings, with less than half their error. Eight successes in eight doses say less about reliability than they might seem to: they bound the probability of a dose inside the band only from below, at 0.69, and no overshoot in eight bounds its probability from above at 0.31 (one-sided 95% Clopper–Pearson limits). Demonstrating a success rate of 0.95 would take 59 doses without a failure.

The best single dose of the campaign did not repeat. The screening run `corner-09` had dosed in 52 s with a 0.7 mg error, but eight more doses at its settings took 47–385 s (median 202 s), depending on how much powder happened to fall after the bulk stage stopped (22–69 mg). The observed Pareto front therefore overstated what that setting could do, and the recommendation based on the model's averaged prediction held up where the single best dose did not.

Most of the time saved came from the tap stage (Fig. 3c). A hand-tuned dose spent 152–318 s, 84–93% of its time, tapping out the last 54–66 mg at 0.6–1.1 mg per tap, whereas at the recommended settings each tap freed 1.5–3.6 mg, a difference consistent with the tube rising from a 10° trickle tilt to a 15° tap tilt instead of dropping to it. The optimizer changed seven of the eight settings, and all six numeric settings ended at an edge of their search range (Fig. 3d). Some of these edges are hardware limits, such as the auger speed near its maximum, and others are choices, such as the 3 mg stopping band; the rest suggest that the next campaign should search a wider range.

The salt settings also transferred to a different powder. Applied unchanged to AlSi10Mg, with the afterflow constant fitted on salt, they delivered 7.993 g for an 8 g request (−0.09%) in 315 s, in a single dose that has not yet been repeated. An unsieved Al 4047 batch from our atomizer, in contrast, clogged the auger on coarse agglomerates. Free-flowing powders thus appear to share

much of their calibration, while powders that flow differently need their own, which is the situation a multi-task model is designed for.

3.3. Multi-task calibration across powders

[This subsection reports results that do not exist yet; the text below shows the intended form.] With data from [three] earlier powders, the multi-task model reached 90% of the best known hypervolume on a held-out powder in [N_{MT}] doses, against [N_{ST}] for single-task optimization and [N_{Sobol}] for quasi-random search (Fig. 4a). Carrying over the best settings of the most similar powder gave a good first dose, without any dose on the new powder, for free-flowing powders but [failed for cohesive ones]. The learned correlations between powders (Fig. 4b) [grouped the free-flowing metal and salt powders together and separated the cohesive and hygroscopic ones], and the descriptor kernel built from dosing fingerprints [matched the index kernel after one dose on the new powder]. In the prospective test on a freshly atomized batch, the recommended settings [dosed 0.5 g in X s with Y mg median error after Z doses]. For the accessible-dose-range objective, the smallest target dosed within tolerance with probability 0.95 was [50 mg for AlSi10Mg and 200 mg for salt].

3.4. A discrete element model of the auger

Without any fitted parameter, the discrete element model reproduced the delivery of salt at the campaign's centre-point settings: 105 mg per revolution at 27.5° and 60 rpm averaged over the run (116 ± 20 mg after the first quarter revolution), against 105 ± 11 mg measured over eight doses (Fig. 5a). A second random packing at the same settings gave 102 mg. The model also reproduced the rise of delivery with tilt, but with larger residuals at the other settings: 33 against about 28 mg per revolution at 0° (+18%), 153 against about 195 mg at 45° (−22%), and 71 against 106–110 mg at 22.5° and 90 rpm (−33 to −35%). The measured values at 0° and 45° were taken at 30 rpm and scaled to 60 rpm with the measured speed exponent.

The model is most sensitive to the geometry of the exit and to the grains. The tip of the auger core sits inside the exit and leaves a 1.07 mm annular gap, about 2.5 grain diameters, which restricts the flow: moving the tip back by 0.5, 1 and 2 mm raised delivery from 105 to 112, 224 and 427 mg per revolution. Reducing the median grain size from 0.425 to 0.35 mm raised it by 29%. In a smaller 10 mm auger, adding cohesion (20 kJ m^{-3}) cut delivery from 46.8 to 0.2 mg per revolution, the same switch from flow to no flow seen

between the two silicon grades. The model's cost sets what can be simulated. The outlet section with 29,000 grains takes 26–30 min per revolution on one processor core, a full salt-filled auger with 1.1 million grains about 19 h, and a fine metal powder would need about 10^9 grains per auger, so those powders require coarse-graining [70].

[Inference results.] The emulator built from [N] simulations reproduced held-out simulations within [X%]. Delivery depended most strongly on cohesion at low tilt and on friction at high tilt (Fig. 5b), so the tilt sweep of the dosing fingerprint constrains both. For salt the posterior gave an effective cohesion energy density of [X kJ m^{-3}] and sliding friction of [Y] (Fig. 5c); for [the other powders] the values are listed in Table [S1].

3.5. Dosing response and powder properties

Even with literature values in place of measured properties, the dosing response ranked powders the way standard flow measures do (Fig. 6a). Across 12 powders, delivery at 45° fell as the angle of repose rose (Spearman $\rho = -0.84$, bootstrap 95% interval -0.98 to -0.50; $p = 0.0006$, or 0.023 after a Bonferroni correction for the 42 pairs examined), and the same held at 22.5° ($\rho = -0.77$, $n = 11$). The five powders with a measured or manufacturer angle of repose ranked correctly on their own: those at 30–32° delivered 244–339 mg per revolution and those at 41–45° delivered 9–37 mg. The equilibrium moisture content predicted the error of 1 g doses under the hand-tuned controller ($\rho = +0.87$, $n = 9$): every powder dosed within 5 mg held less than 0.1 wt% water, and every hygroscopic biopolymer missed by 33–292 mg. Literature median particle size did not predict delivery across materials ($\rho = +0.26$), although it decided flow within one material, as the silicon grades show. The shape of the tilt response also differed between powders (Fig. 6b), which is the information the discrete element model needs to separate cohesion from friction. The interruption fraction, computed from the same balance records, ranged from 0.13 for AlSi10Mg and 0.26 for salt to 0.81 for carboxymethyl cellulose and 0.96 for sodium alginate. It also followed the angle of repose ($\rho = +0.76$, $n = 10$) while only partly overlapping the volumetric flow index ($\rho = -0.66$ between the two), so the two descriptors are not redundant.

[Comparison with measured properties.] On the same lots, the effective cohesion inferred from dosing correlated with the shear-cell flow function coefficient ($\rho =$ [X], Fig. 6c) and with the Hall-flow time of the powders that passed the funnel ([Y]); [Z] powders did not pass the Hall funnel, but all of

them produced a dosing fingerprint. Against blade-spread single layers, [the interruption fraction ranked the batches like layer coverage].

4. Discussion

Calibration data as a flow measurement. The doser measures a powder under conditions close to those of its use: a small sample moving through a narrow gap under gravity, shear and vibration. Every quantity in the dosing fingerprint is collected during routine calibration, and the same measurement works inside a glove box, where Hall funnels and shear cells are awkward to use. The full set of protocols takes 4–10 g and under an hour; when the calibration has to be done anyway, storing and analysing its records adds little. The fingerprint has two further advantages over funnel tests. It still returns information on cohesive powders: where nothing moves, a defined protocol records a no-flow result rather than a failed test, and wherever powder moves at all it returns tilt, tap and afterflow descriptors, whereas a funnel that does not flow returns only a failure [18, 77]. And it has several dimensions, so it can separate effects that a single flow time combines. The correlation with the angle of repose, and the failure of particle size to predict delivery across materials (Section 3.5), agree with studies of pharmaceutical screw feeders, in which bulk density and cohesion-related properties, rather than size, predicted feed rate [26, 27, 28, 78]. The literature values used here are class estimates, however, not measurements of our lots, and a correlation of ranks across 12 very different materials shows only that the doser orders powders sensibly. Whether it can resolve the smaller differences between batches of one alloy, which matter most for feedstock quality control in AM, remains to be shown [(Section 3.5)].

What Bayesian optimization contributes. On salt, the optimizer found settings three times faster than hand tuning, with less error, in one unattended session of under 2.5 h. Three lessons from the campaign shape the multi-powder design. First, repeatability has to be an objective, not an afterthought. The fastest dose on the observed front came from settings whose time varied eightfold between repeats, because the powder that falls after the auger stops varies from dose to dose. A noise-aware acquisition function [41], a recommendation based on the model's averaged prediction, and a validation block together guarded against this, but an explicit repeatability objective, estimated from repeated doses, would let the optimizer trade it against speed

directly. Second, a short budget finds a good setting but does not prove it best. The 14 optimizer doses never beat the best single screening doses, which owed part of their result to chance, and only the validation block showed which setting was both fast and repeatable. Time per dose also had a long tail of slow doses (the eight doses at the centre point took 24–391 s), so a noise model that allows for occasional very slow doses, and a longer budget, would sharpen the comparison. Third, the recommended settings sat at the edges of the search space, so the ranges were set too conservatively; this is easy to fix but should be expected for every new powder. The 20 screening doses were well spent: they explained, in physical terms, why taps during the bulk stage speed doses up but cause overshoots, and they gave the model a balanced starting set. Because overshoot is irreversible, the current formulation's symmetric absolute error should become a one-sided penalty with a constraint on the probability of overshoot, as the formulation for the four-objective campaigns does (Section 2.5).

Sharing calibration across powders. That the salt settings dosed AlSi10Mg to within 0.1%, in a single transfer check, suggests that powders which flow freely share much of their calibration, and that a multi-task model can start a new free-flowing powder close to its optimum. Cohesive and hygroscopic powders, which deliver an order of magnitude less per revolution and respond differently to tilt and speed, are less likely to benefit from the salt data, and a model that learns task correlations decides this from data rather than by assumption. Using the dosing fingerprint to compute the task correlations would also let the model place a new powder before its first optimized dose [45]. [Results of the multi-task benchmark (Section 3.3) to be discussed here.]

From dosing data to contact parameters. The discrete element model reproduced salt delivery at one setting without calibration and followed its trend with tilt, which suggests that its geometry and contact laws capture the main mechanism. Its errors at other settings, and its strong sensitivity to the exit gap and the grain size, mean that inference must start from a measured exit geometry and a measured particle-size distribution; otherwise those uncertainties will be absorbed into the inferred cohesion and friction [52, 59]. The inferred parameters will also be effective rather than intrinsic: with a softened Young's modulus, the cohesive force in the SJKR model depends on the contact area and hence on the chosen stiffness, so inferred cohesion energies can be compared across powders only at a fixed numerical stiffness [50]. Static

(triboelectric) charging, suspected in the noisy readings of carboxymethyl cellulose and seen as salt sticking to the auger walls in a glove box, is not represented in the model [79]. If these caveats are respected, the same contact parameters can drive DEM models of powder spreading, which are the standard tool for predicting layer quality in powder bed fusion [60, 61, 15, 62].

Limits of the present data. All doses so far were made in air, on one auger geometry, and the optimization covers one powder at one target mass. Barium chloride caked after humidity cycling, and the correlation of dose error with moisture content shows that ambient humidity matters for hygroscopic powders. The tap stage sets the cost of accuracy. All 16 validation doses landed inside the 3 mg band, but taps took most of each dose's time, and the mass a tap frees falls as the lip of the tube empties (on salt, from about 2.8 mg for the first tap after the auger moved to below 0.5 mg after 40 taps). Tighter tolerances or faster doses therefore need a smaller and more consistent final increment, for example a smaller outlet or a refill of the lip between taps, more than a different control law [64]. The feed factor also drifted between sessions, by about 10% for salt between two days and up to 1.8-fold across sessions with different fill levels. A fingerprint therefore needs a standardized powder state (fill level, priming and conditioning) and a same-day reference powder before small differences between batches can be read from it. Finally, every powder was supplied commercially or made in our laboratory without a recorded lot analysis, which is why measured properties of each lot are part of the work that remains.

Implications for additive manufacturing. In an alloy-discovery workflow the doser is calibrated on every powder anyway, so the records described here come at little extra cost in powder or time. Stored with the dose records, the fingerprint, the validated settings and the inferred contact parameters for each lot form a feedstock record that can flag a batch that flows unlike its predecessors, as the clogged Al 4047 batch would have been flagged before it reached a printer. [Whether that record predicts spreading and packing in the powder bed will be tested on blade-spread layers (Section 2.7).]

5. Conclusions

We treated the calibration of an open-source auger doser as multi-objective Bayesian optimization and asked what the calibration data say about the powder. On salt, an unattended campaign of 42 doses found settings that

dosed 0.5 g in a median 78 s with a median error of 1.8 mg over eight repeats, three times faster than hand-tuned settings, and those settings dosed 8 g of AlSi10Mg to within 0.1% without change in a single transfer check. At fixed settings, 13 powders spanned a 24-fold range in delivery per revolution, three did not flow at all, and delivery ranked the powders like their angle of repose. A discrete element model of the auger, with no parameters fitted to dosing data, reproduced the measured delivery of salt at one setting and its trend with tilt. [Multi-task calibration across N powders reduced the doses needed for a new powder by X%, and effective cohesion and friction inferred from dosing data tracked shear-cell and Hall-flow measurements.] Because every dosing workflow already calibrates its doser, the records of that calibration can double as a characterization of each feedstock at little extra cost.

CRediT authorship contribution statement

[To be completed once the author list is agreed.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data, code, CAD files and simulation inputs are available at `https://github.com/vertical-cloud-lab/powder-doser` [and will be archived with a DOI on acceptance].

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Code (Anthropic) in order to draft and revise the text and to write the figure scripts, and Edison Scientific in order to search the literature and to obtain

mock reviews of the draft. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Table 2: Descriptors of the dosing fingerprint. "Measured" means available from existing records for the number of powders given; "exploratory" means computed once from existing records with a protocol that has not been repeated; "planned" means not yet measured.

Descriptor	Unit	Source	Replicates	Status
Delivery per revolution at 0, 22.5, 45°	mg/rev	protocol C, 30 rpm	6 rev per tilt	measured, 13
Revolution-to-revolution scatter (CoV)	%	protocol C	6 rev per tilt	measured, 10
Speed exponent α	–	protocol D, 15/45/90 rpm	3 rev per speed	measured, 8
Mass freed by one tap	mg/tap	protocol E	8 taps	measured where resolved, 7
Afterflow time constant τ_{af}	s	stops in closed-loop doses	all stops of a run	measured, 6
Interruption fraction	–	15 rpm balance increments	1 run	exploratory, 10
Volumetric flow index	–	45° delivery / bulk density	20 rev	planned

Table 3: Search space of the salt campaign (target 0.5 g). The two tap settings are categorical and the rest continuous. The hand-tuned settings are those found by hand for this controller on salt before the campaign; the recommended settings are dose `bo-005`.

Setting	Range	Hand-tuned	Recommended
Taps during bulk stage	off / 2 Hz	off	2 Hz
Taps during trickle stage	off / 2 Hz	off	off
Bulk tilt (°)	15–40	30	40
Bulk speed (rpm)	20–100	55	100
Switch to trickle, mass to go (g)	0.05–0.30	0.25	0.30
Trickle tilt (°)	10–30	15	10
Tap tilt (°)	0–15	10	15
Stopping band $\pm\delta$ (mg)	3–15	5	3

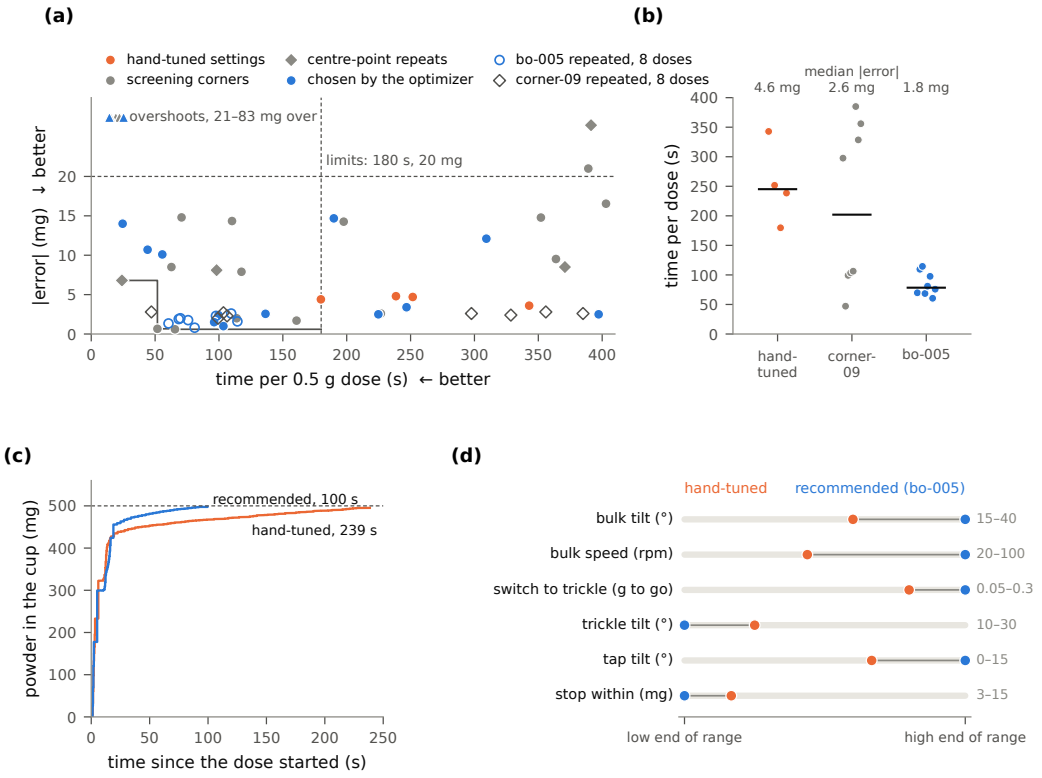


Figure 3: Multi-objective Bayesian calibration of the doser on salt (target 0.5 g). (a) Time and absolute error of every dose; triangles at the top mark overshoots beyond the 20 mg limit, the dashed lines are the limits that define the hypervolume, and the step line is the observed Pareto front. Open markers are the two validation blocks of eight doses. (b) Time per dose for the hand-tuned settings (4 doses) and the two validated settings (8 doses each); bars are medians, and the numbers above give the median absolute error. (c) Powder in the cup during a hand-tuned dose and a dose at the recommended settings. (d) The six numeric settings, each drawn on its own search range, for the hand-tuned and recommended settings.

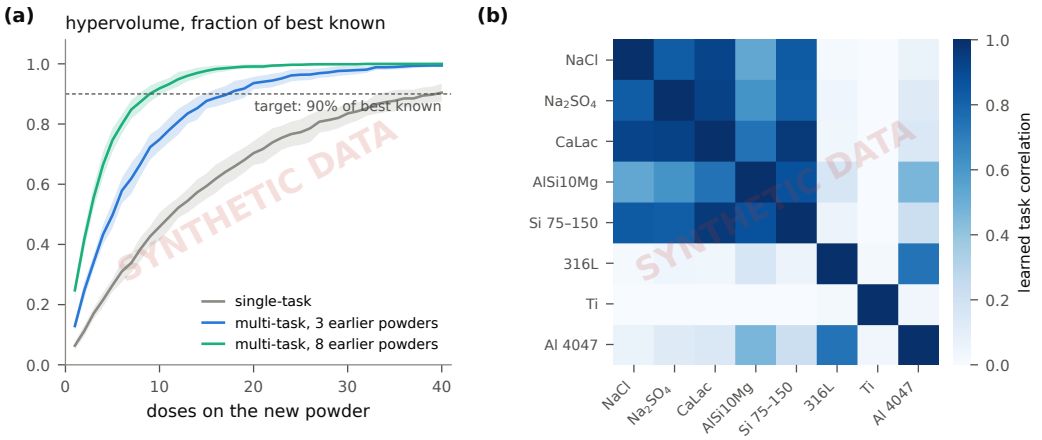


Figure 4: [Placeholder: synthetic data.] Multi-task calibration across powders. (a) Hypervolume reached on a held-out powder against the number of doses on that powder, for single-task optimization and for multi-task optimization informed by three or eight earlier powders (mean ± one standard deviation over ten replays). (b) Correlations between powders learned by the multi-task model. All numbers in this figure are invented to show the form of the result.

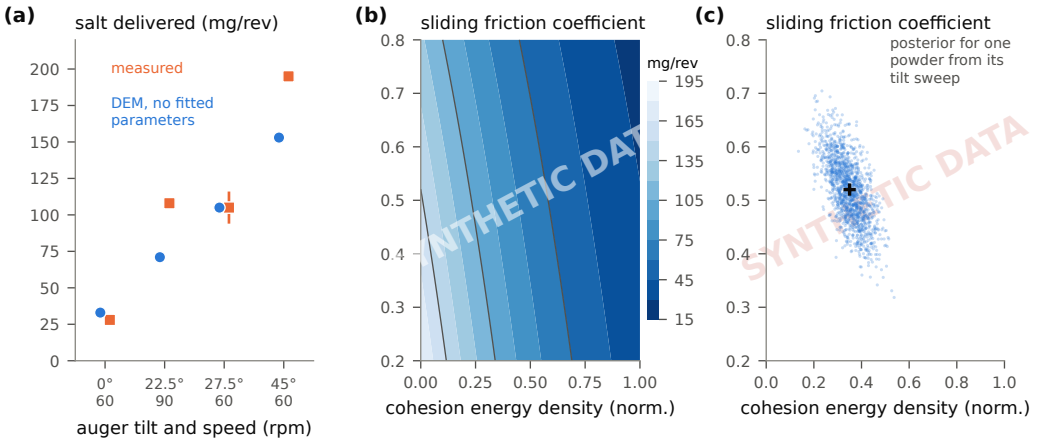


Figure 5: Discrete element model of the auger. (a) Salt delivered per revolution, simulated with no parameters fitted to dosing data and measured on the rig, at four combinations of tilt and speed. At 27.5° and 60 rpm the measurement is the campaign's centre point (eight doses, bar: ± one standard deviation); at 22.5° both simulation and measurement are at 90 rpm (bar: range of runs); the 0° and 45° measurements were made at 30 rpm and scaled to the simulated 60 rpm with the measured speed exponent. (b) [Placeholder: synthetic data.] Delivery per revolution against effective cohesion and friction from the emulator. (c) [Placeholder: synthetic data.] Posterior distribution of the two parameters for one powder given its tilt sweep. Panels (b) and (c) show invented numbers to illustrate the form of the result.

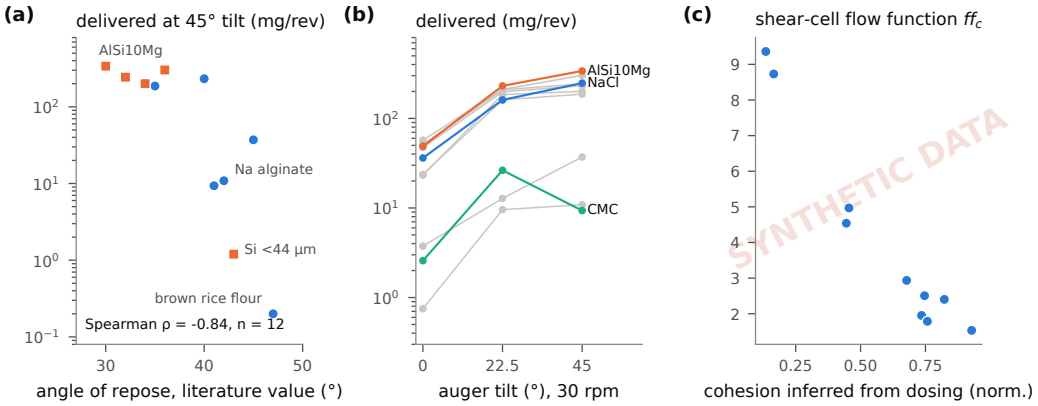


Figure 6: Dosing response against powder properties. (a) Delivery per revolution at 45° tilt and 30 rpm against the literature angle of repose for the material class (seven of twelve values are engineering estimates); the finer silicon grade (one 7 mg pulse in six revolutions, 1.2 mg per revolution) and brown rice flour (0.2 mg per revolution) are plotted at their means, which are not resolved from zero. Blue circles: food-safe stand-ins; orange squares: alloy-relevant powders. (b) Delivery against tilt for the ten conveying powders; three are highlighted. (c) [Placeholder: synthetic data.] Effective cohesion inferred from dosing against the shear-cell flow function coefficient measured on the same lots.